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Requirements and Characteristics for the Development and Selection of Design Methods

Johannes Matschewsky *, Sergio A. Brambila-Macias, Abhijna Neramballi and Tomohiko Sakao *

Division of Environmental Technology and Management, Department of Management and Engineering, Linköping University, 581 83 Linköping, Sweden; abhijna.neramballi@liu.se (A.N.)

* Correspondence: johannes.matschewsky@liu.se (J.M.); tomohiko.sakao@liu.se (T.S.); Tel.: +46-(0)-13-28-1635 (J.M.)

Abstract: While many design methods are developed, tested and reported in the literature, their utilization in industry practice remains low. Design methods are receiving substantial scholarly focus and are considered central to efficiently achieving reliable outcomes in the engineering design process. They are particularly vital as industrial companies increasingly transition to integrated offerings of products and services with a lifecycle perspective, leading to additional uncertainty and complexity. Thus, the presented research aims to support method selection and development, focusing on resource-efficient offerings. This is achieved through an in-depth, practice-centric, empirical study of users' requirements of design methods and the corresponding characteristics of design methods aimed at meeting these requirements in resource-efficient offerings. Highly relevant insight supporting a broad set of stakeholders is reported. Firstly, the user requirements and method characteristics reported support practitioners seeking to identify a design method fitting their needs. Secondly, academics and practitioners aiming to enhance the usefulness and impact of a design method may benefit from considering these requirements and characteristics during method development. Lastly, the systematic approach taken in this research can be applied by both method developers and potential users to identify additional requirements and corresponding characteristics specific to their conditions. Two use cases for the results attained are reported, focusing on applying the research results for method selection and deriving overall guidelines for developing design methods directed toward resource-efficient offerings.

Keywords: engineering design; design method selection; user requirements; method characteristics; resource efficiency; circular economy

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1. Introduction

Design methods are recognized as crucial support for design in industrial practice [1,2], e.g., to provide transparency and traceability [3,4] and to reduce complexity and uncertainty early on in the design process [5]. Today, many companies are expanding their value capture by moving towards integrated offerings of products and services [6]. In the wake of this expansion and the associated uncertainty, a particular need for method support for both the transition to and the design of these offerings has arisen [7]. Due to the broadness of business models and concepts described in this context [8–10], this research uses the term *resource-efficient offerings* as an accumulative description of the currently diverse terminology of offerings focusing on more efficient resource use throughout the lifecycle, as also used in, e.g., [11].

While many methods with varying complexity and aims have been presented to support the design, deployment and operation of resource-efficient offerings [12,13], the industrial uptake of these methods is subject to discussion [14–16]. This is also true of engineering design at large [17–19]. Several challenges to the industrial adoption of methods have been reported, e.g., not having a fitting format for industry deployment [20],

insufficient fit with established processes [21] and lacking customizability [22]. Remarkably, the implementation and use of existing design methods are rarely reported and reflected upon in academia, where many of these methods originate [2]. However, when design methods are applied in practice, they often originate in industry rather than academia [23].

At this point, there is a lack of academic knowledge about how the selection and development of design methods can be improved to enhance their use in industry. This concerns both engineering design in general [24] and resource-efficient offerings in particular [25]. The gap manifests itself particularly in the lack of a structured understanding of both the requirements of design method users and characteristics indicating whether a design method is fit to fulfill these requirements. Approaching this gap is both relevant and timely, as new insight on requirements for method development has emerged [19], and researchers continue to create and spread repositories of design methods [26]. In a meta-analysis by Cantamessa [27], as referenced in Jagtap et al. [28], in 47% of the cases, method developers fail to disclose how the method they are proposing was developed and which assumptions underpinned its development. Thus, missing support for transparent and traceable method selection and goal-oriented method development may be central reasons behind the lackluster use of these methods, whether by industry practitioners [29] or academics.

In response to this challenge, the research presented here has the following main objective: *to support the development and selection of design methods for resource-efficient offerings through empirically based and practically evaluated user requirements and method characteristics*.

To this end, through a structured process applying in-depth literature reviews, semi-structured interviews, an extensive survey and multiple workshops involving 40 practitioners, 12 key requirements and five key characteristics (with two qualifiers each) were identified. The respondents to the empirical research are all active in resource-efficient offerings, although the results may be relevant for engineering design in general.

By presenting both the results of the research as well as the research method in detail, the authors further aim to contribute to a central question in engineering design research: Reflexiveness. As Reich [2] points out: "[. . .] We [design researchers] are also designers and, therefore, users of design methods. It would be expected of us [. . .] that we specifically use our methods in our own practice. [. . .] There are hardly any documents of such decision processes". To facilitate a push towards more reflexive research on design methods, the attained results and the systematic approach are intentionally focused on industrial method designers and users and academic method developers and users [28,30]. By providing a detailed account of the research method, the data collected and the lessons learned from a large number of industry practitioners and academics, the authors aim to enhance the potential of this article to contribute to much-needed reflexiveness in the field [31,32].

The empirical results are applied in two illustrative use cases. Use case A demonstrates the assessment of existing methods concerning the characteristics. This illustrates the utility of the method characteristics to analyze existing methods to attain a structured, traceable basis for method selection. Use case B departs from the knowledge of a method developer. Based on a correlation assessment of user requirements and method characteristics, guidelines for method development in the expert's field of expertise were derived. This use case illustrates the applications of the requirements and characteristics in the early phase of method development to determine the central properties of a to-be-designed design method to increase its usefulness in practice.

The research reported makes the following contributions to academia and practice: The identified user requirements towards design methods provide critical insight for both practitioners and academics seeking to tailor the solutions they develop towards the needs of their future users. The reported method characteristics support analyzing existing methods and defining target marks for design method development, which may be relevant for practitioners and academics alike. Further, replicating the research method enables method developers in industry and academia to isolate additional, tailored requirements,

characteristics and their potential correlations relevant to contexts going beyond the design of resource-efficient offerings.

2. Research Background

In this section, the background of the research is introduced. Based on this, the research gap and motivation are identified.

2.1. The Value of Design Methods

Before discussing how the selection and development of methods could be supported, it must be made clear why they are needed. Overall, researchers in engineering design agree on the importance of methods for theory and practice. Concerning their importance for design research, methods are regarded as a cornerstone [33] as academics endeavor to support practitioners in design processes since *"research in design has focused on the development of systematic methods in the pursuit of providing designers with generalizable and universal instructions for design"* [34]. Lindahl [35,36] describes methods as an important support for companies in coping with an accelerating need to adapt to changes continually.

Concerning their expected practical impact, design methods play an essential role in design processes and activities [22,37] in reducing errors, shortening development time and increasing the quality of the design result [22,38,39]. Additionally, reducing complexity and uncertainty in the design process is a crucial function of design methods [1,40]. From a more general point of view, methods are seen as a means to help designers achieve the desired outcome as efficiently and effectively as possible [34,41,42]. Frey et al. [43] focus on the challenge of proving the effectiveness of a method either through a mathematical model or assessing its real-life performance, as indicated by case studies and practitioner experience. Kannengiesser and Gero [44] have found overlaps of design processes in engineering design, software design and service design, suggesting that lessons learned in different design fields may also have broad relevance. Lastly, Gericke et al. [45] point out the educational value of design methods in educating students as an additional meaningful point of assessing their value.

2.2. General Criticism and Identified Causes of Low Method Use and Ways Forward

A concern often voiced in prior research is that only a small portion of the methods produced in academia enjoy wide use in practice [18,22,46]. Reasons for this include failed method deployments in industry environments [39] and a general unawareness of the possible benefits of using design methods [18]. Further, critical causes of low design method utilization are lack of time, purchase cost and cost of education and knowledge, all pointing to a lack of management commitment to highly efficient design [36].

Wolf [47] identified management support as a critical prerequisite for successful method adoption and use, as is the availability of the information necessary for practitioners to make informed decisions concerning their challenges [48]. A further challenge identified is that overly ambitious implementation efforts often exceed the absorptive capacity of companies for change in their processes, as pointed out by Wallace [49], referring to Birkhofer et al. [29]. Researchers have also identified challenges with adapted and adjusted design methods in industry use [50], leading to insufficient results and dissatisfaction. Lofthouse [51] points out the frustration of many users with workshop-style methods, as they are time-consuming and often stand in contrast to practitioners' day-to-day practice.

In focusing on the methods as such, researchers identified several challenges and shortcomings. A key challenge is the time-consuming nature of many methods and the method users' lack of awareness that methods may be adjustable to the available time and resources [20,36,52]. At times, practitioners fail to see the usefulness of a design method in their daily work altogether [36]. Another critical aspect is that methods are often presented in a format not optimized for use by practitioners—e.g., in the form of lengthy text descriptions and manuals. In reference to this, Geis et al. [22] point out the practitioners' need for methods "focused on output and less theoretical ballast". Similarly,

Jänsch and Birkhofer [46] and O'Hare [53] note that academically produced methods are often presented abstractly and overly scientific. More recently, however, researchers have lamented a perceived predominant focus on communicating and teaching methods rather than improving their value for practitioners [19].

A key point mentioned by Knight and Jenkins [21] is the need of methods to fit with established processes and the ability to integrate with existing structures [22]. Vasantha et al. [13] postulate a need for precise methods fit to solve individual issues, with Tukker [9] arguing against such an approach, fearing that very tailored approaches lack universal applicability. Similar conclusions are reached by Geis et al. [22] and Le Pochat et al. [54], emphasizing the need for adaptability and customizability of design methods. Lofthouse [51] points out a growing need for methods to be able to handle low-quality or inconsistent data. Particularly, methods used to facilitate the adoption of a new business model may need to provide such flexibility. Concerning the question of specificity versus strategy orientation, Birch et al. [55] make a case for specific methods and tools to achieve optimal performance.

Recently, in-depth work on the *Theory of Design Methods* has emerged. Daalhuizen and Cash [56] proposed *Method Content Theory*. Departing from this, Cash et al. [57] pointed out the lack of a consolidated procedure for assessing design methods and, in response, proposed a systematic assessment framework to evaluate the efficacy and effectiveness of design methods. While these contributions add to our knowledge with measurable characteristics of design methods and on evaluation of design methods, these are generalizations based on theoretically prescribed properties of "good theories" and "good artifacts".

Currently, there is a lack of a practice-centric perspective on which characteristics of design methods should be considered by method developers to ensure their effectiveness among practicing design professionals. Furthermore, there is a lack of a mechanism to select design methods with specific characteristics that align with the contextual needs of the method users.

2.3. Particularities for Adoption and Use with a Focus on Methods for Resource-Efficient Offerings

Particular challenges apply where methods concentrating on resource-efficient offerings are concerned: Where methods focusing on engineering design often concentrate on achieving the optimal technical solutions (see [58]), methods focused on resource-efficient offerings are applied in environments of even greater uncertainty [59,60]. The focus is extended from the design of physical objects to the design of products and services [61,62], expanding even to the design of parts of business models [63,64] and a strong focus on holistic value generation for providers and customers [65]. The broadness of the offerings and the resulting complexity may also lead to confusion and low adaptation of academically conceived supports, even in this field [66]. A method or tool focusing on resource-efficient offerings should be adapted to the company culture, as reported by Lindahl [67], a fact exacerbated by the challenges of changing company culture when adjusting to providing resource-efficient offerings [7,68]. Watz and Hallstedt [69] go the way of lifting out a single topic, the integration of sustainability into the product design process, as a way of circumventing the lack of adoption of a design method and still achieving the intended impact.

2.4. Research Gap and Motivation

While the value of design methods in industry practice is well-described, and many methods to support designers in practice are made available, very few enjoy broad adoption and use in industrial practice. The causes for this situation have been investigated, and steps towards enhancing the communication and utility of such methods have been taken in prior research. However, an in-depth investigation with a dual emphasis on both design method users' requirements and the design methods' characteristics to fulfill these has not yet been conducted. This is particularly critical in resource-efficient offerings, a new area of business industrial companies increasingly venture into, making method support particularly valuable to cope with the complexity and uncertainty that accompany the transition. Practitioners aiming to select a method are currently not provided with empirically based

support to analyze both the needs of users and the capabilities of methods to fulfill these needs. Further, academic and industry-based method developers lack support in method development for high utility. The method requirements and characteristics are intended to support both the development of new methods and the reflexive assessment of existing methods in this respect.

3. Research Method

This section outlines the methodical approach used to conduct the research presented in this article. The research method applied supported the identification, evaluation and correlation of information regarding both user requirements for design methods and design method characteristics. Two use cases focus on method selection and method development to investigate the relevance and applicability of the characteristics and requirements identified.

3.1. Identification of Practitioner Requirements

Industrial method user requirements were identified through four consecutive activities: A literature review, expert interviews, follow-up meetings and a web-based survey. Each of these activities is further detailed in the following subsections.

3.1.1. Literature Review

The literature search on user requirements for design methods started with a search focusing on product-service systems, design, lifecycle, engineering, sustainability, resource-efficiency and environment carried out with the Web of Science database. Of the initial 1254 research items found in 2016, after screening for document type (article or review, 598 items), abstract screening (150 items) and content analysis, 49 papers were found to be relevant to requirements on design methods for resource-efficient offerings. A narrative literature review [70] complemented this search based on well-known sources on design methods such as Ulrich and Eppinger [24], Unger and Eppinger [71] and Dieter and Schmidt [72]. Further, research that emerged during the review process was added. The review database was updated throughout, and relevant references were added.

3.1.2. Interview Study with Industry Practitioners

The literature-based findings were formulated into questions for interviews with practitioners in companies based in Sweden. This interview guide provided the background for the semi-structured interviews carried out, focusing on the existing challenges of method identification, implementation and efficiency and effectiveness in use. All face-to-face interviews took place in 2016. The interviewees were selected according to their experience with product and service design. While participants' job positions varied from owners of small companies to quality managers and designers in large companies, for simplicity and relevance to this research, their roles were divided into product design, service design or both. This was determined by examining the respondents' typical working tasks. Where it was artifact-centric, the field of expertise was identified as product design. Where the focus was on the use phase and services provided to maintain and support the artifact, the field of expertise was determined to be product design. If individuals were involved in both areas, both fields were included. Table 1 provides an overview of the interviews conducted.

Table 1. Interviews conducted to identify user requirements for design methods for resource-efficient offerings.

No. of Interviews	Industry Sector	Interviewee's Field of Expertise
5	Floor grinding and cleaning	Product design (2), Service design (3)
2	Remanufacturing of electronics	Product design (1), Service design (1)
3	Sustainable materials for the construction and paper mill industries	Product and service design (1), Product design (2)
1	Facade cleaning	Service design
1	Sales and installation services of engineering equipment	Service design
4	Transportation	Product design (3), Service design (1)
5	Heavy duty and off-road vehicles	Product design (4), Service design (1)
3	Health care products	Product design (2), Service design (1)

Follow-up meetings were held with practitioners from all companies, and the minutes were used to validate or extend the requirements identified. This was also an opportunity to receive feedback from the companies involved.

3.2. Evaluation of the Most Relevant Requirements through a Practitioner Survey

To identify the relative importance of the user requirements on design methods for resource-efficient offerings identified through literature review and interviews, a survey was answered by 25 practitioners from eleven companies. Three additional companies from the eight involved in the interview study were added to the survey for greater depth. The web-based survey was live from August 2017 to March 2018. A Likert scale from 0 (not important at all) to 5 (extremely important) and an additional option for "I do not know" were used.

3.3. Identification of Method Characteristics

Method characteristics were derived to operationalize the industry requirements on design methods and to simplify their application to both the development of design support and the assessment of existing design methods. Initially, these characteristics were motivated by challenges to method adoption and use in the industry identified in a narrative literature review [70] presented in an earlier conference publication [73] and a questionnaire conducted with industry practitioners at a large company (ibid.). Based on this, in multiple brainstorming sessions [74] involving all authors, an initial set of characteristics was discussed and adjusted. This led to the derivation of two qualifiers for each characteristic to further clarify said characteristics' intent and facilitate their application. These qualifiers have three key properties: (a) semi-quantitative measurability; (b) allow both positive and negative tendencies regarding a given requirement; and (c) relevance for the process of developing a design method and the use by an industry practitioner.

3.4. Evaluation and Correlation—Focus Group Methodology

3.4.1. Workshop Methodology

Workshops were conducted in a focus group format [75]. The format of focus groups was altered from reliance on an interview guide and a strong moderator role [76] to focus on a data-gathering sheet available to all participants in advance. Based on a QFD-style approach [77], this document comprised both the survey results focusing on method requirements and key method characteristics as evaluation parameters. Here, the goal was to establish correlations between requirements and characteristics. Ideally, this would, e.g., indicate which method characteristics a practitioner with certain requirements should focus on when selecting a design method. The scale for the correlations given was 0 (none), 1 (low), 2 (medium), 5 (high). Before the workshops, this document was tested by one author individually and by another as a pair and was supplied to the academic expert panel four weeks before the workshop.

The assessment of the correlations between method characteristics and method user requirements was intended as a central aspect of the workshop with academic method developers. However, this structure quickly made way for a more interactive discussion, which is characteristic of focus group sessions. The emphasis shifted towards reasoning and discussion between the participants, particularly about the method characteristics, and data were gained by the participants' mutual interaction and reaction to one another during the session [78].

The workshops were recorded and stored digitally. The digital recordings were analyzed and selectively transcribed [79]. From this record, aspects central to the results attained and the discussion based on these were identified. These structured outcomes of the workshops mostly focused on the relevance and comprehensiveness of the characteristics, providing the basis for the discussion of the research results and the summarized characteristics presented.

3.4.2. Academic Panel

The academic panel was comprised of specialists in the design of resource-efficient offerings, particularly in developing design methods and testing and implementing these in various environments. The researchers have been carrying out these activities in Brazil, Denmark, Japan, the Netherlands, Sweden and other regions, working with companies ranging from small enterprises to large multinational businesses. Table 2 shows the participants' fields of research and experience.

Table 2. Academic panel assessing method characteristics and their relation to designers' requirements.

Alias	Position	Years in Research and Practice on Resource-Efficient Offerings at Time of Workshop	Details	Total Citations (April 2024 Google Scholar)
Expert A	Full Professor	17	Professor in system design, experience with designing and deploying methods in various fields of business and companies of varying size	6700+
Expert B	Full Professor	15	Professor in industrial design with a focus on engineering design, concentrated on sustainable engineering in the packaging sector	2400+
Expert C	Associate Professor	10	Extensive experience in eco-design and PSS design research and practice, founding partner of an eco-design consultancy with globally active clients	7600+

3.4.3. Industry Practitioners

Survey respondents were invited to participate in a panel discussion. Practitioner A has over a decade of experience in leading the transition of product-oriented businesses towards lifecycle-oriented product-service integration, both at a large OEM in the logistics sector and consulting. Practitioner B has extensive experience working with lifecycle assessment during the design stage in the automotive sector, both directly with an OEM and in consulting. Table 3 condenses information on the participating practitioners.

Table 3. Practitioner panel discussing method characteristics and their impact in practice.

Name	Position and Work with Design Methods	Industry	Years Worked with Design Methods
Practitioner A	Core technology developer, focused on business transformation towards integrated offerings of products and services with lifecycle perspective; has introduced and developed design methods in various areas of business.	OEM in the logistics sector	10+
Practitioner B	Environmental specialist, focused on analyzing the impacts that products and services have on the environment.	Automotive and public sectors	10+

4. Results: Method User Requirements, Characteristics and Their Application on Resource-Efficient Offerings

4.1. User Requirements on Design Methods for Resource-Efficient Offerings

Based on the approach described in Section 3.1, 20 requirements were identified. A comprehensive overview of the literature reviewed, as well as detailed interview results, can be found in a project report [80].

The survey results showed that the respondents ranked twelve of the 20 requirements highly. Table 4 presents these twelve most relevant requirements and the literature supporting them. Statistical data regarding the practitioners' evaluation can be found in the Supplementary Materials for this article (see Table S1) and a project report [81].

Table 4. User requirements on design methods for resource-efficient offerings and supporting literature.

Category	Requirements	Supporting Literature
Outcome	Useful in early design phase Provide quantified results	[24,61,71,72,82–84]
Process	Support prioritization of improvement areas in design Support communication across relevant actors Support collaboration across relevant actors	[24,58,85–93]
Content	Support lifecycle thinking	[12,83,94–106]
Time and cost	Low cost of usage	[36,67,107–112]
Users	Easy to use Easy to learn Support designers' creativity Provide learning opportunity Support management of multiple projects	[2,13,36,67,71,88,102,113–117]

4.2. Design Method Characteristics for Resource-Efficient Offerings

This subsection reports the results of the focus group workshops concerning method requirements, characteristics and qualifiers providing additional depth to the characteristics with academic method designers and industrial method users.

An initial set of characteristics was derived from a literature review and prior empirical work [73]. This initial set can be found in the Supplementary Materials for this article (see Table S2) to provide the reader with a comprehensive view. The workshops with academics and industry practitioners notably impacted the resulting characteristics and qualifiers. Central feedback received and the most important discussion points and quotes from both workshops are laid out in the Supplementary Materials (Document S3) to provide full transparency.

Table 5 provides a comprehensive overview of the resulting characteristics, the qualifiers operationalizing them and their properties.

Table 5. Characteristics and qualifiers for developing and selecting design methods for resource-efficient offerings.

Characteristic	Description	Qualifier	Properties
Customizability—Understanding interfacing and modularity	• Concerned with the adaptability of methods to circumstances in industrial practice. • Can be key where extensive methodologies are considered impractical or where process integrity is vital.	Modularization	Provisions taken within a design method to be split into at least two parts with interfaces of in- and output between the two.
		Adjustability to existing process	Provision of effective interfaces by the design method to connect to the established processes in an industrial environment, facilitating the in- and output of information to and from the design method.
Simplicity—Understanding input and process	• Main concern lies in the extensiveness of the method process and the data required for its execution.	Simplicity of data addressed	Refers to the dimensions and the uncertainty of the data handled by the design method.
		Simplicity of entailed process	Describes the number of steps, iterations and conditions evaluated by the method process.
Flexibility—Understanding data, knowledge and usage modes	• Focused both on the conditions the method is applied in and the in- and outputs of the method.	Input flexibility	Describes the design method's capability of accepting different types and quality of data. This can range from very rigorous all the way to very flexible requirements.
		User flexibility	Refers to the knowledge, skill and experience required of the user to successfully execute a design method. Linked to this, this also refers to whether a design support can be carried out by a qualified individual, or requires a pooling of knowledge, skill and experience in a workshop-style environment.

Table 5. *Cont.*

Characteristic	Description	Qualifier	Properties
Scalability—Understanding scope and lifecycle	• Refers to the area in focus of the design support's assessment. • Particularly critical in the design of highly integrated solutions based on complex systems of products and services with high provider responsibility throughout the lifecycle.	System scale	Refers to the focal range of the design support with respect to the total solution to be delivered. This may vary between an encompassing approach focusing on the entirety of the solution to a very specific method focused on only a small part of the resource-efficient offering.
		Time scale	Concentrates on the design support's area of focus in terms of lifecycle phases. Thus, methods can range from a focus on individual lifecycle phases to design supports focusing on the entire lifecycle.
Clarity—Understanding communication and useAl	• Concentrates on the way a design support is communicated to its prospective users as well as its ease of use and systems in place to aid its implementation and application in an industrial environment.	Form of implementation	Describes the availability of an analog (paper-based data-collection document) or digital tool to help the user carry out the method.
		Communication	Refers to the way the method is conveyed to the user by analog (e.g., academic paper, handbook) or digital means (e.g., screencast, video tutorial).
Utility—Understanding outcome recipient and process responsibility	• Focused on how the method caters to the needs of the recipient of the outcome of the design method's use. • Also concentrates on the ability of interested parties to follow up on decisions made during method use and to attain an understanding of such processes.	Recipient	Describes the method's transparency on what function or individual is the intended recipient of the method outcome and how the needs of the intended recipient are considered.
		Traceability and accountability	Refers to the method's capability to make clear the individual decisions made in a method process and their motivation, as well as the individual or group involved in the decision-making process.

4.3. Use Case A—Method Selection: Assessment of Existing Methods for Design of Resource-Efficient Offerings

4.3.1. Use Case Description

Three methods enjoying broad use in the field of resource-efficient offerings are evaluated against the characteristics, illustrating a case of method selection. The method characteristics weighted relative to the user requirements on design methods are applied. Expert A completed the weighting during the QFD-style correlational assessment of requirements and characteristics as part of a workshop (see Section 3.4.1).

This approach is intended to indicate how a method selection could occur. However, the methods are not interchangeable, have different foci and are not compared focusing on a particular industry challenge. Besides demonstrating the evaluation of an existing method using the characteristics identified, the use case shows how successful methods can exhibit very different characteristics depending on their use in practice.

The assessment only focuses on positive correlations, meaning that all relations assigned between a characteristic and a method signify that the method in question possesses the respective characteristic, resulting in a binary assessment. Negative correlations, meaning that the *absence* of a particular characteristic (e.g., modularity) can be desirable at times, require a clear use case to be meaningful.

The following design methods used in resource-efficient offerings and prevalent in academia and industry are evaluated. Life Cycle Assessment (LCA) is a well-known technique used to assess a product's potential environmental impacts throughout its life cycle [118]. It is used for decision-making in industries and governmental or non-governmental organizations. Failure Mode and Effect Analysis (FMEA), initially developed by the US military, is widely used by practitioners for failure analysis of product or service systems [119]. Based on this method, Environmental Effects Analysis (EEA) was developed, which identifies and evaluates the environmental impacts of a product during its early stages of development [120]. This provides the opportunity to choose alternative materials and processes during the early stages (ibid). QFD is a widely used design tool applicable in the early stages of product development [77]; QFD for environment (QFDE) is an extension of the tool that incorporates environmental requirements and supports Ecodesign [121].

4.3.2. Use Case Results

These methods are semi-qualitatively assessed in Table 6 to examine whether they consider the method characteristics and the corresponding qualifiers established in Section 4.2. Earlier assessments of these methods in relation to different method parameters [122] were used as points of departure.

The stark differences in how the design methods correspond to the characteristics are immediately apparent: LCA is a very structured but thereby also inflexible method, requiring, for example, clearly defined datasets and substantial user knowledge, and is often ill-advised for use directly in the design process [123]. In contrast, EEA is much more flexible, explicitly focusing on semi-quantitative assessments, which can also be based on low-quality data. The same can be said about QFDE. Further, EEA and QFDE are flexible regarding input data and can be carried out relatively easily. However, LCA does offer comprehensive and rigorous analysis from a life cycle perspective, thus satisfying the "time scale" qualifier of the characteristic "scalability." It satisfies both the qualifiers of the characteristic "clarity" as it can utilize several existing digital tools [124] to analyze and communicate the results, similar to EEA and QFDE.

4.3.3. Use Case Discussion and Lessons Learned

The assessment reinforced the need for an explicit goal at the outset of the design method selection. Expert A performed a correlational assessment of method characteristics and user requirements. A non-specialist interpreting Table 6 as a comparison of methods for a similar use case may falsely conclude that QFDE is the best of the three methods analyzed since it corresponds to 87% of the weighted characteristics assessed. Even when leaving out LCA because of its very different nature and concentrating on similar methods, EEA and QFDE, such an assessment would be misleading. Therefore, it is critical to clarify that a high level of compliance of a given method with the identified characteristics is not indicative of a better method. Therefore, assessing methods concerning the identified characteristics is only meaningful when a use case is defined, and the user understands which characteristics are important for the method to support said use case.

Overall, a downside of the assessment is its extensiveness and time-consuming nature, particularly when including the QFD assessment leading to the weighting of the qualifiers for the respective method characteristics. Further, the individual/s carrying out the assessment must have substantial knowledge of the evaluated methods, potentially making parts of the evaluation obsolete. The subjectivity of the assessment carried out is a further limitation. Thus, although the semi-quantitative evaluation of methods against weighted characteristics must be handled carefully, the overall assessment demonstrated how a method selection can occur in an industry environment. Further, the assessment showed which characteristics drive the success of the methods when considering their use in industrial practice. For LCA, a restrictive structure and low flexibility for data sources are vital to attaining reliable results that withstand external and internal scrutiny. EEA and QFDE are successful because they can be flexibly adapted to individual problems, and users can quickly learn how to apply them.

Assessments of methods with varying foci may also prove insightful when working with engineering design students. Getting a deeper understanding of the characteristics of design methods and their varying foci may sensitize students to the particularities of their use. This, in turn, may lead to those students effectively and efficiently using design methods in practice in the future.

Overall, this use case indicates that method selection in design for resource-efficient offerings is supported and structured by the requirements and characteristics identified. A user may gain confidence and traceability concerning the decision made, and possible deficiencies in the selected method are pointed out and can be addressed before implementation.

Table 6. Assessment of existing methods (LCA, EEA and QFDE) and the extent to which they consider the identified method characteristics.

		Characteristics												
		Customizab.		Simplicity		Flexibility		Scalability		Clarity		Utility		
	Qualifier	Modular-ization	Adjusta-bility	Data	Process	Input	User	System	Time	Implemen-tation	Comm-unication	Recipi-ent	Trace-Ability	Rel. Σ
	Weight	0.06	0.09	0.11	0.13	0.14	0.13	0.14	0.11	0.08	0.01			1
Method	LCA	1	0	0	0	0	0	1	1	1	1	0	0	0.40
	EEA	0	0	1	1	1	0	0	1	1	1	1	1	0.58
	QFDE	1	1	1	1	1	0	1	1	1	1	1	1	0.87

4.4. Use Case B—Support for the Design of Resource-Efficient Offerings

4.4.1. Use Case Description

This case focuses on identifying lessons learned for developing a design method. The context of these lessons learned is determined by industry requirements and Expert C's knowledge and background. Expert C has worked in the aerospace industry as a design engineer and, as an academic, has extensive experience closely collaborating with the industry on implementing ecodesign, product-service systems and other resource-efficient offerings, publishing seminal papers on these issues. Thus, the expert has a unique set of knowledge and experience stemming from both industry and academia, indicating a strong relevance of the collected data.

This departs from the correlation assessment between the method user requirements and the method characteristics carried out by Expert C during the workshop, as detailed in Section 3.4.1. The weights assigned to the user requirements by industry practitioners as a result of the survey are used with the correlation assessment to arrive at ranked and weighted method characteristics.

Based on this result, guidelines are formulated. In a follow-up step, these guidelines may support another individual's development of a design method in an industry context. To this end, the prioritization of the method characteristics and particularly remarkable correlations between method characteristics and method user requirements were evaluated.

4.4.2. Use Case Results

In order to provide full insight, both the weights assigned by 25 design method users in industry and the correlational assessment carried out by Expert C are shown in Table 7. The weights given for user requirements are the practitioner assessments shown in Table S1 normalized to a 0–1 scale.

Table 7. Results of correlation between weighted requirements and characteristics.

			Characteristics									
			Customizability		Simplicity		Flexibility		Scalability		Clarity	
		Weight	Modularization	Adjustability	Data	Process	Input	User	System	Time	Implementation	Communication
Requirements	Useful in early design phase	0.76	1	2	5	1	5	1	1	2	0	0
	Provide quantified results	0.63	1	0	2	0	5	1	1	1	1	0
	Support standardizing design process	0.6	5	5	1	5	0	1	0	0	0	0
	Support prioritization of improvement areas in design	0.73	0	0	2	0	1	1	2	2	1	1
	Support communication across relevant actors	0.77	0	2	2	2	2	5	1	1	1	1
	Support collaboration across relevant actors	0.71	2	5	2	1	1	5	1	1	1	1
	Support lifecycle thinking	0.81	1	1	1	1	2	1	2	5	1	1
	Easy to learn	0.83	1	2	1	2	1	1	1	1	1	5
	Easy to use	0.88	2	5	0	5	2	1	1	1	2	2
	Support designers' creativity	0.84	1	1	1	1	1	1	1	1	1	1
	Provide learning opportunity	0.64	1	1	1	1	1	2	1	1	1	1
	Support management of multiple projects	0.68	5	2	1	2	1	1	1	1	2	1
	Weighed ∑		14.1	19.3	13.9	15.7	16.3	15.4	9.8	13.0	9.1	11.1
	Normalized ∑		0.10	0.14	0.10	0.11	0.12	0.11	0.07	0.09	0.07	0.08
	Rank		5	1	6	3	2	4	9	7	10	8

Overall, in the eyes of Expert C., a focus on allowing the method to be designed to the individual circumstances in industrial practice appears to be most critical in fulfilling the requirements of industry users. Adjustability to existing company processes and a

method's capability to accept different *types* and *qualities* of data appear to be most central, as is a *simple method process*.

According to Expert C, whether certain requirements are met by the method to be designed appears to rely mainly on one or very few characteristics and qualifiers. Customizability is the critical characteristic that determines whether a design method can support managing multiple projects. However, none of the characteristics appear to impact the learning value gained from using the method, nor do they support the designer's creativity. Clearly, how easily a method can be learned is primarily determined by how it is communicated, as the characteristic clarity and, particularly, the qualifier communication show a high correlation to the easy-to-learn requirement.

In contrast, the requirement *support communication across relevant actors* is supported by a broad set of characteristics, particularly *simplicity* and *flexibility*. Yet, for a method to fulfill the requirement *useful in the early design phase*, almost all characteristics and qualifiers play a role. At the same time, a focus on simplicity and flexibility is indicated.

4.4.3. Reflexive Use Case Discussion and Lessons Learned

First, correlating requirements and characteristics through a QFD-style approach must be discussed. The results in Table 7 show that the sums for the scored characteristics may be considered low relevance, as they cover a very small portion of the available scale. However, taking a closer look at individual correlations between requirements and characteristics does reveal interesting insights, which are the basis for the lessons learned below.

When examining the case results, Expert C's background in the development and industry deployment of academic design methods becomes apparent: The most highly ranked qualifiers are focused on the adjustability and modularity of the method to be designed. This is natural, as academically conceived methods may be used in various circumstances.

Below, some deductions are presented as lessons learned based on Expert C's assessment in the context of the given industry requirements. The requirements in focus are shown in italic print, while the characteristics and qualifiers mentioned are added in brackets for clarity. These lessons learned may be partly transferable to other contexts, depending on different industry needs:

- To ensure a method supports a standardized design process and can be integrated with existing design processes, academic method designers can focus on developing modular and adjustable methods (customizability: modularization adjustability).
- Ensuring the design method developed is useful early in the design process requires the ability to accept little and rough data and still provide a meaningful result (simplicity: data, flexibility: input).
- To ensure a method supports communication across relevant actors, it can be helpful to ensure it is adjustable to existing processes and can be used by a broad set of practitioners with broad backgrounds (flexibility: user).
- If an essential requirement on a design method is its ease of use, the adjustability to existing processes and the simplicity of the method should be considered (customizability: adjustability, simplicity: process).
- When being easy to learn is a critical requirement, it appears that how the method is communicated is more important than the method's content. Therefore, providing a handbook or video instruction can be helpful (clarity: communication).

If *supporting a designer's creativity* and *providing a learning opportunity* are important goals, field testing cannot be circumvented as these topics appear too complex to be covered by the identified characteristics.

5. Discussion—Enhancing Design Method Development, Selection, Implementation and Use

A central criticism in prior research investigating the low adoption rate of design methods is their unsatisfactory deployment in industrial environments [39]. As industry practitioners remarked, the proposed method characteristics and the requirements identi-

fied support an informed and goal-oriented method selection. This may increase the chance of successfully deploying a new method to alleviate a practical challenge [48].

As criticism of a predominant research focus on the challenge of marketing and disseminating methods instead of focusing on their utility is being raised [19], the presented dual emphasis on method characteristics and method user requirements may fill a critical gap. Where the method user requirements allow scrutiny of the challenge to be solved and the needs of the people involved in solving it, the characteristics and qualifiers point to properties to ensure a selected or newly developed method corresponds to these requirements. Further, by assessing a method focusing on the characteristics, an understanding of the properties a method does and does not have is provided.

When focusing on delivering a solution for a challenge under the restriction of a given absorptive capacity in a company [49], the additional information provided regarding the properties of methods after examining them using the presented results can lead to more reliable decision-making. This is expected to increase the likelihood of selecting or developing a method that can solve the challenge at hand and be usable in daily practice. In an area such as resource-efficient offerings, marked by a high level of uncertainty [59,60], this support may be of particular value to enable a successful design process.

Much prior research was directed to how academically created design methods are structured, communicated and used [36,50,52]. Based on the results reported, design method selection and development are streamlined and traceable for parties not part of the process. Although more time is spent early on, the implementation and use of the design method eventually selected may be accelerated, leading to overall time savings and additional learning opportunities [36].

Considering these points, where do the results presented fit into the ongoing theory-centric discussion on design research and the impact of academically conceived design methods? The expert participants of the academic workshop agreed the results support a structured and goal-oriented method selection and development process, improving the academic status quo. Thereby, a step has been taken toward much-needed clarity in this area of design research [19]. Further, a practice-centric dimension is added to the theory development on method properties [56] and the proposals made therein. The proposed requirements and characteristics may further provide added depth to method evaluations carried out according to Cash et al.'s [57] framework.

Discussing the requirements and characteristics proposed when publishing new design methods allows for reflexiveness during the research process [31]. Besides, while the development process of a design method was formerly often a black box [27,28], basing its decision-making on well-understood requirements and characteristics indicates a shift toward much-needed transparency. It allows scrutiny of the process and building on others' results transparently and traceably. Insight published after the data collection for this article points in a similar direction: "*An articulation of the intended use [of the design method] will ease evaluation, as the target audience, required rigor, allowed adaptations and expected benefits would be clearly defined. Evaluation against more clearly defined objectives would rely less on interpretation*" [125].

6. Conclusions

While a large number of design methods are developed and made available by academics, few appear to be used in industrial practice. The research presented aimed to propose highly relevant user requirements on design methods and design method characteristics fit to fulfill these requirements in resource-efficient offerings. This dual emphasis on method user requirements and corresponding method characteristics should enable industry users to identify fitting methods efficiently. At the same time, both academic and industrial readers are supported in developing methods tailored to real-world challenges in a traceable way.

Based on extensive empirical work incorporating interviews, surveys and focus group workshops with industry practitioners and academic method designers, the results re-

ported ensure users can easily identify critical requirements and characteristics for method development and method selection, supporting traceability and clarity in this process. Figure 1 summarizes the identified requirements and characteristics.

The method characteristics allow for a performance review of method selection and development and adjusting existing methods based on identified shortcomings. Further, by also focusing on academic method developers, the results support reflexive, rigorous and impactful design research practice.

Resource-efficient offerings are, for many companies, a new area of business involving high uncertainty. These companies, in particular, require support from methods to facilitate the novel offerings' design, deployment and operation. To this end, the requirements and characteristics focused on this area represent a step towards effective and efficient design of such offerings. While all data-gathering was carried out with practitioners and academics in this area, the identified requirements and characteristics may be broadly relevant for method development and selection in engineering design at large. This is worth exploring for both practitioners and researchers.

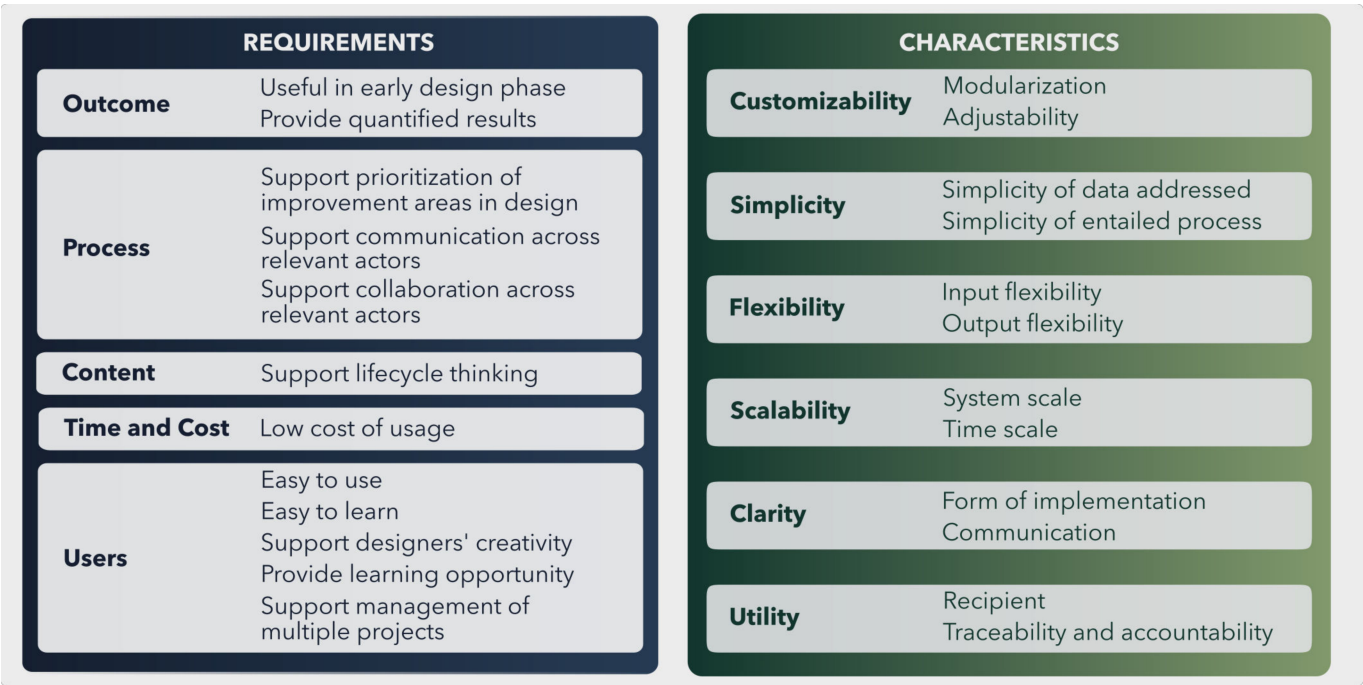


Figure 1. Summary of the identified design method requirements and characteristics.

In future research, identified requirements and characteristics should be applied and tested in developing and evaluating design methods for resource-efficient offerings in research projects, with industry practitioners and with engineering students. Further, the scope of design objects could be expanded to additional contexts, such as business model design. To continue the advance to more reflexive research in engineering design, the research community is encouraged to test and expand upon the results presented as part of the continued discussion on design methods toward the goal of highly effective and efficient design methods in industry use.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/designs8030059/s1, Table S1. User requirements on design methods for resource-efficient offerings, supporting literature and their relative importance based on practitioner evaluation. References cited therein: [2,12,13,24,36,58,61,67,71,72,82–110,112–117]. Table S2. Characteristics and qualifiers for the development and selection of design methods for resource-efficient offerings before the workshops. References cited therein: [9,13,20–22,29,46,49–55,67,114,120,126–129]. Document S3. Quotes from different stakeholders involved in the study.

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